

- We propose **CGIL**, a new way to adapt CLIP to a Continual Learning scenario.
- **CGIL** combines prompt-learning techniques with latent generative replay.
- We introduce a new metric (CI-Transfer) to assess zero-shot performance on unseen classes.

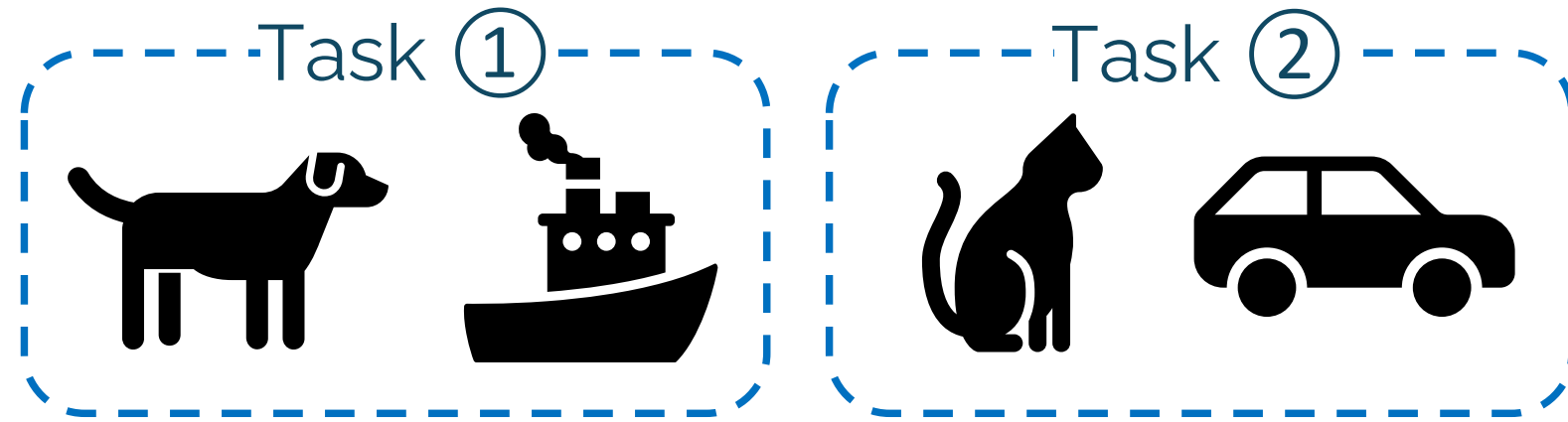
TL;DR

Motivation

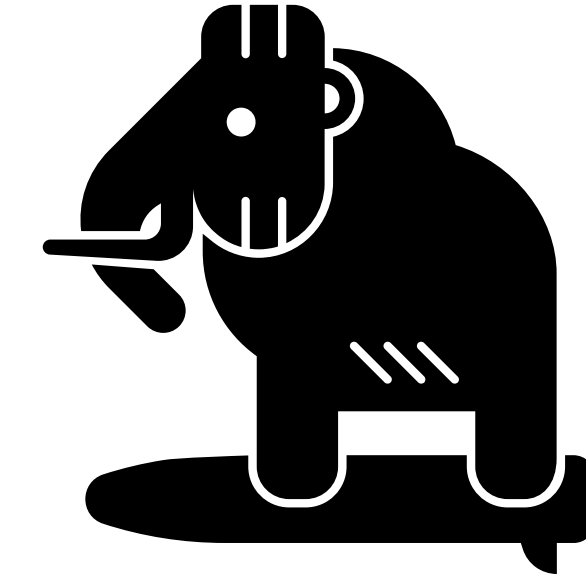
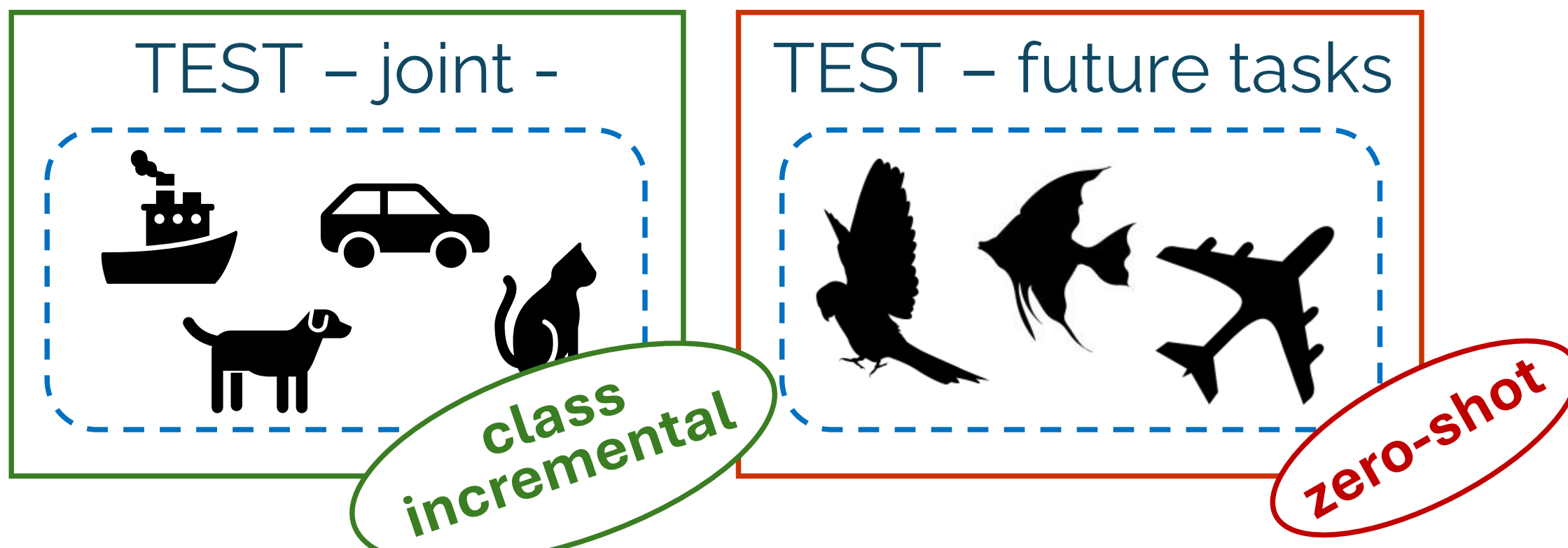


Continual Learning (CL) aims at training models that **retain past knowledge** while **learning new tasks**, thus preventing Catastrophic Forgetting.

TRAIN – sequential tasks –



TEST – joint –



Mammoth

A PyTorch benchmark library for **Continual Learning** with 20+ datasets and 50+ methods.
aimagelab/mammoth



We aim to incrementally fine-tune CLIP while enhancing its **zero-shot performance** by **leveraging inter-domain knowledge** shared between previous and future tasks.

Continual Generative training for Incremental-Learning

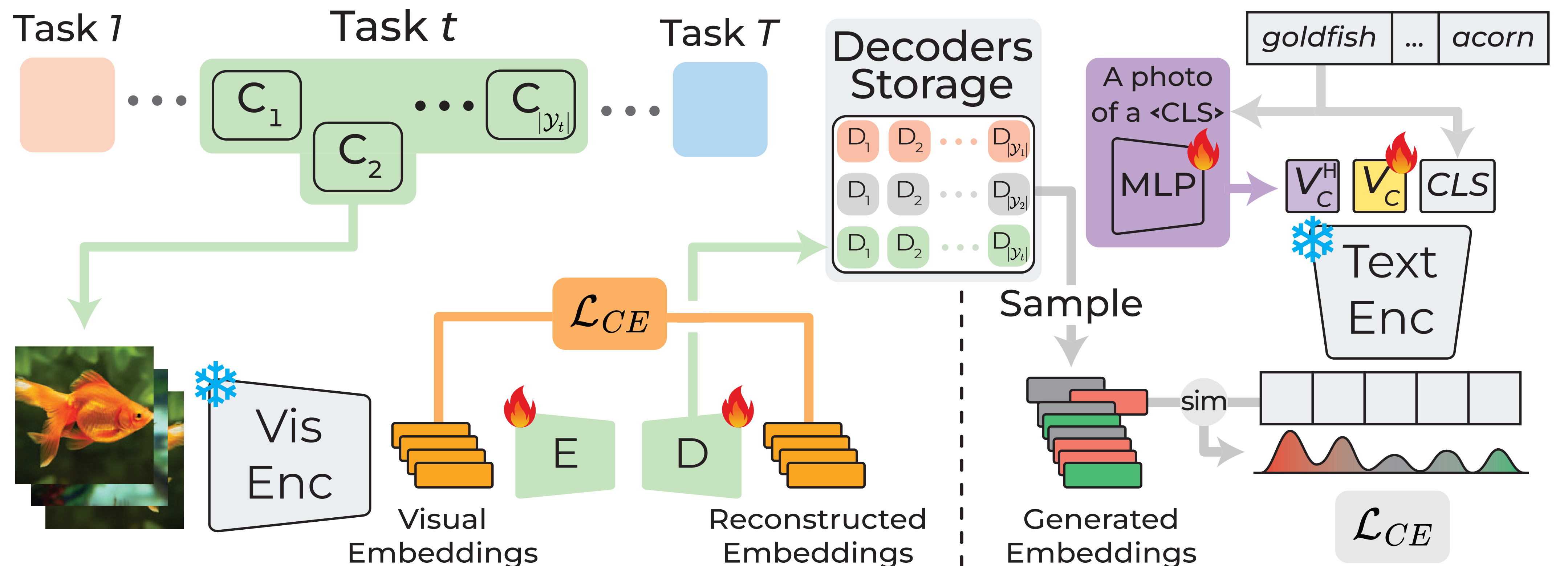
For each task:

Phase 1: Learning CLIP latent embedding distributions of input images with VAEs.

Store the VAE decoders.

Phase 2: Align *past and current* class-specific context prompts from *only* generated embeddings.

Inference: Employ trained prompts for seen classes and zero-shot CLIP embeddings for unknown classes.



Class Incremental Learning

We evaluate **CGIL** across a wide range of datasets with different domains.

Final Acc.	Img-R	Cars-196	CUB-200	EuroSAT	ISIC	Avg.
Zero-shot CLIP	81.95	64.99	50.52	53.32	26.59	55.47
LwF	19.09	23.24	16.73	25.13	33.06	23.45
GDumb	44.28	28.74	61.34	90.99	61.64	57.40
DER++	56.66	53.66	74.62	<u>93.08</u>	<u>65.68</u>	68.74
L2P	66.49	38.18	62.21	46.34	47.13	52.07
DualPrompt	68.50	40.14	66.00	71.39	49.99	59.20
CODA-Prompt	75.45	31.99	67.30	63.12	44.87	56.55
AttriCLIP	87.39	75.63	58.28	72.33	28.26	64.38
SLCA	77.00	67.73	84.71	88.69	59.19	<u>75.46</u>
ZSCL	89.14	77.66	62.43	79.13	34.14	68.50
MoE Adapters	90.67	<u>77.76</u>	64.98	80.56	34.52	69.70
CGIL	<u>89.42</u>	89.27	<u>83.12</u>	96.17	73.03	86.20

Algorithm: given datasets D_t , $t \in 1, \dots, T$

$\mathcal{M} \leftarrow \{\}$ $\triangleright$ initialize a memory buffer for storing the VAE decoders

for each dataset D_t , $t \in 1, \dots, T$ **do** $\triangleright$ iterate over tasks

Learning class-specific VAEs in latent space

for each class $y \in \mathcal{Y}_t$ **do**

$\mathcal{V}_t^{(y)} := \{E_{vis}(x_i) \mid x_i \in \mathcal{D}_t^{(y)}, y_i = y\}$ $\triangleright$ CLIP visual embeddings

Train a new **VAE** = $\{E_y(\cdot), D_y(\cdot)\}$ on $\mathcal{V}_t^{(y)}$

$\mathcal{M} \leftarrow \mathcal{M} \cup \{D_y(\cdot)\}$ $\triangleright$ store the decoder

end for

Prompt alignment through generative replay

$\mathcal{S} = \{(\hat{z}_{vis}, c) \mid \hat{z}_{vis} \sim D_c(\cdot), \forall c \in \mathcal{Y}_1, \dots, \mathcal{Y}_t\}$ $\triangleright$ synthetic dataset

Construct **prompts** $\mathbf{t}_c \forall c \in \mathcal{Y}_1, \dots, \mathcal{Y}_t$

$\mathbf{t}_c \leftarrow \mathbf{t}_c - lr \cdot \nabla_{\mathbf{t}_c} \mathcal{L}_{GR}$ $\triangleright$ optimize the classification loss

end for

Ablations

	Img-R	Cars-196	CUB-200	EuroSAT	ISIC	Avg.
CoOp (<i>Joint</i>)	89.24	89.89	82.52	96.25	73.57	86.29
CoOp (<i>Fine-tune</i>)	84.94	68.61	59.84	79.27	37.08	65.95
CGIL	89.42	89.27	83.12	96.17	73.03	86.20
Different generative approaches						
Multinomial Gaussian	83.89	82.59	80.06	85.70	51.89	76.83
Mixture of Gaussians	88.54	88.82	82.10	93.04	62.42	82.98
Diffusion Models	89.28	90.14	83.48	95.73	68.99	85.52
Different techniques to create the context prompt						
Class-specific token	89.09	88.96	83.06	95.59	72.21	85.78
MLP-generated token	89.41	88.91	82.82	95.69	70.79	85.52
Multiple shared tokens	89.02	88.08	81.50	95.47	70.18	84.85

Zero-Shot over the Future

Let A_t^i be the CIL accuracy on the i -th task after being trained until task $t \in \{0, 1, \dots, T\}$, the **Class-Incremental Transfer** is defined as:

$$\text{CI-Transfer} = \frac{1}{T-1} \sum_{t=1}^{T-1} \left(\frac{1}{T-t} \sum_{i=t+1}^T A_t^i \right)$$

CI-Transfer	Img-R	Cars-196	CUB-200	EuroSAT	ISIC	Avg.
Zero-shot CLIP	82.14	66.16	50.86	55.00	22.42	55.32
AttriCLIP	85.75	73.98	54.07	59.69	24.14	59.53
ZSCL	85.30	72.49	<u>62.76</u>	<u>69.74</u>	<u>25.31</u>	<u>63.12</u>
MoE Adapters	88.25	<u>75.82</u>	61.73	55.77	21.06	60.53
CGIL	<u>86.71</u>	78.80	66.34	71.52	48.18	70.31